Reward engineering, simulator fidelity and lap-time benchmarks

RL driving agent in OCD: Barcelona and Shanghai

Ibrahim Khaled

2 November 2026

Outline

1. Reward engineering: from line tracking to a time-optimal reward
2. Simulator check: AV-21 leftovers in the “F1” car
3. Verification: Limebeer-style optimal control (OCP) reproduced
4. Results: Barcelona vs. Limebeer, Shanghai vs. real world and QSS
5. Open points and next steps

Reward engineering

Old reward (`ref_tracking`, run `full1`)

Per 50 Hz tick, e = lateral offset to the reference line ξ_{ref}:

$$r = a\,\Delta s \;-\; b\,\Delta t\,\big(1+\underbrace{\dot{u}^{\top} W\,\dot{u}}_{\text{jitter}}\big) \;-\; \underbrace{c\,e^{2}}_{\text{tracking}} \qquad a = b = 1,\; c = 0.1,\; W = \mathrm{diag}(0.01, 0.001, 0.001)$$

Failure (off track, stall, wrong way): $P = 50 + 30\,c\,e_{\text{exit}}^{2}$.

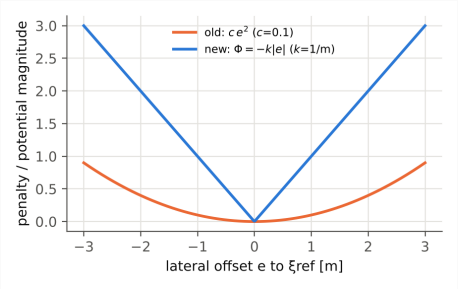
What we saw (Shanghai, AV-21 car, full1 at 2.95M steps):

- Driver glued to ξ_{ref}: mean $|e| = 0.39$ m, max 1.83 m.
- Lap cost split: time 123.3, **tracking 175.9**, smoothness 10.4. Tracking was a larger term than lap time.
- Best clean lap 122.85 s. The reference line is a *safe* min-curvature line, so following it closely means driving slowly.

New reward (`time_pbrs`, R1): what changed

$$r = a\,\Delta s - b\,\Delta t \;+\; \big[\gamma\,\Phi(s') - \Phi(s)\big], \qquad \Phi(s) = -k\,|e(s)|,\; k = 1\,\mathrm{m}^{-1},\; \gamma = 0.992$$

1. **Jitter term removed.** $\dot{u}^\top W \dot{u}$ is no longer rewarded (only logged).
2. **Line penalty linear, not squared:** $k|e|$ instead of $c\,e^2$.
3. **Line term is potential-based shaping** (Ng et al. 1999), so the optimal policy is the same as with no line term at all. ξ_{ref} only guides early learning.
4. Constant failure penalty $P = 100$; a stall truncates the episode instead of ending it.



Why these two changes

Removing the jitter (smoothness) term

- Top racing agents (Linesight/Trackmania, GT Sophy) use no smoothness term in the reward.
- Christmann et al. (IROS 2024): a smoothness penalty can make the agent settle for worse performance, and its weight is hard to tune.
- The time cost already penalises wasted inputs. Smoothness stays a logged diagnostic.

$|e|$ **instead of** e^2

- e^2 is flat near the line and steep far from it. Small deviations cost almost nothing, large ones dominate the return.
- $|e|$ has a constant gradient (Linesight also shapes on a linear distance).
- Inside a PBRS potential the term cannot change the final driver. It only speeds up early learning.

Effect of the new reward

Run	Car / reward	Best clean lap 2	Own QSS (safe line)
full1	AV-21, old reward	122.85 s	133.9 s
r1_warm	AV-21, new reward, warm start	123.04 s	133.9 s
sf70h_mac_r1	sf70h_tum, new reward, from scratch	**103.24 s**	104.78 s
p1_TUM	sf70h_tum + TUM-like setup, new reward	**101.27 s**	103.56 s

- The warm start kept the old line-following habits and gained nothing. From scratch, the free driver **beats the QSS lap on the safe line** by 1.5–2.3 s, because it is no longer tied to that line.
- Cost: lower clean rate. Only 48/194 deterministic evaluations were clean in sf70h_mac_r1.

Simulator check: AV-21 leftovers

The “F1” car was partly an AV-21

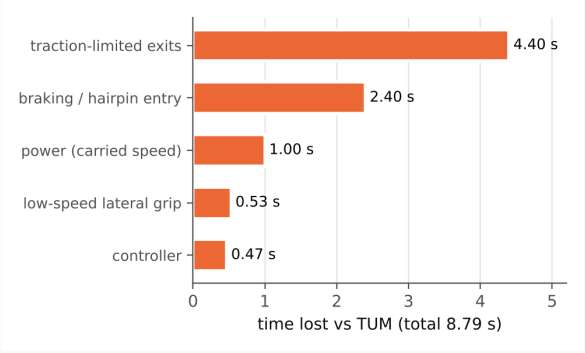
OCD ships the **Dallara AV-21** (Indy Autonomous Challenge car). `sf70h_tum` replaced only the values that TUM's F1 file provides:

From TUM (F1):
mass 733 kg, geometry, peak μ per axle, aero (C_zA 4.88, C_xA 1.56 m^2), 575 kW engine, gear ratios, ERS

Still AV-21 (PLACEHOLDER):
tyre slip stiffness and curve shape (PKX/PKY, PEX/PEY, PCX/PCY), combined slip, inertias (scaled), wheel masses and inertias, tyre vertical stiffness, roll centres, steering lock 0.6 rad, suspension defaults

Consequence: the AV-21 slip stiffness now carries about 30 % more friction, so the rear's longitudinal force peaks at **13–33 % slip**. A real F1 tyre peaks near 8 %. Lateral force has already collapsed by then, so the car cannot use its traction on corner exits.

Diagnosis: TUM's 94.9 s lap driven in OCD



- Closed-loop tracking of TUM's raceline and speed profile in a TUM-aligned OCD car.
- Best driven lap **102.99 s** vs. TUM 94.89 s, while OCD's own QSS is 95.73 s. The *quasi-static* limits match. The time goes in *transients*.
- Longitudinal slip stiffness $\times 1.6$ alone: -2.69 **s**, with QSS unchanged.

Fix: new car `sf70h_f1` (1 Nov)

Item	AV-21 / old	F1 value (source)
Tyre PKX1/2/3	30.7 / 0.27 / 0.13	F 42.17 / −0.96 / 0; R 49.84 / −0.38 / 0 (PL14 peak slip)
PCX1, PCY1; PEX, PEY	1.6, 1.5; AV-21	1.9, 1.9; 0 (PL14)
Steering lock	0.6 rad	0.30 rad
Heave springs	off	1.0 / 0.25 MN/m (F / R)
Brake bias, torque	0.50, 12 kN m	0.58, 15 kN m
Footprint	4.6×2.0 m	5.006×2.07 m (FIA 2017 width)

- Fx peak moves from $\kappa = 0.14$–0.27 to 0.075–0.081. Peak μ is unchanged.
- Closed-loop TUM lap: 102.87 → **101.84 s** (101.29 s with heave damping).
- New finding: with zero heave damping the car porpoises (4.5 Hz heave mode) at high downforce. A heave-damping value is needed.

Verification: Limebeer-style OCP

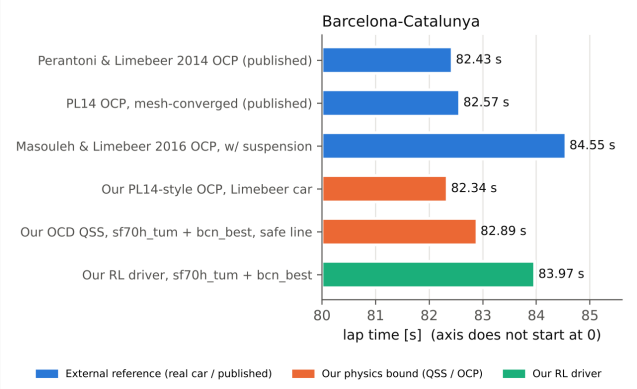
Minimum-lap-time OCP, as in Perantoni & Limebeer 2014

- Planar body, quasi-static load transfer, 4 combined-slip tyres, LSD, aero, power limit. Distance-domain direct collocation (CasADi + IPOPT), mesh down to 0.5 m.
- **D-check, Barcelona with Limebeer's car: 82.34 s** vs. PL14 mesh-converged 82.57 s, i.e. $-0.28\,\%$ (pass band $\pm 1.5\,\%$).
- Mesh-converged within 0.005 s. KKT residuals $< 5 \times 10^{-7}$. IDAS re-integration agrees to < 1 ms in lap time.
- Same model with the SF70H-TUM values: **Shanghai 94.25 s** (TUM QSS 94.89 s), **Barcelona 78.76 s**. With the `sf70h_f1` tyre: 94.62 / 79.38 s.

Limits: local optima only (a different initial guess gave $+0.55$ s). No suspension and no wheel-spin dynamics. This is an external reference, not an OCD lap.

Results

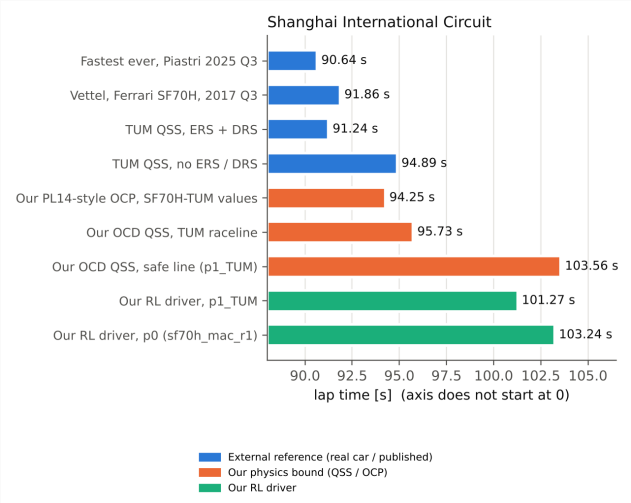
Barcelona: Limebeer's OCP vs. our RL driver



- RL: **83.97 s** (step 5.05M of 7.05M).
- +1.40 s (+1.7 %) to PL14's 82.57 s.
- +1.08 s (+1.3 %) to our own QSS (82.89 s).
- Faster than ML16's suspension model (84.55 s).

Not the same car: PL14 uses a 2013-era F1 car (660 kg, C_L 3.0, 560 kW). Ours is `sf70h_tum` (733 kg, 575 kW) with `bcn_best` at aero_level 1.0 on a PLACEHOLDER aero map. Our OCP with SF70H-TUM values gives 78.76 s.

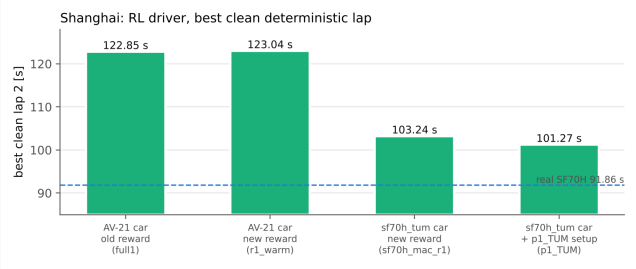
Shanghai: real world vs. QSS vs. our RL driver



No Limebeer lap exists for Shanghai.

- RL **101.27 s**: +9.4 s to Vettel's SF70H (91.86 s), +6.4 s to TUM QSS with no ERS/DRS.
- RL beats our QSS on its own safe line (103.56 s).
- Our QSS on TUM's raceline (95.73 s) is within 0.84 s of TUM.

Shanghai: how the gap closed

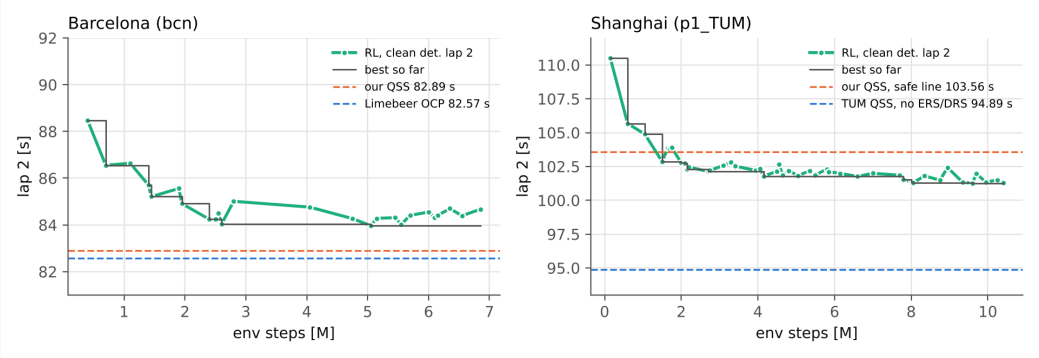


Gap to the real SF70H: **31.0 s** → **9.4 s**.

Remaining gap (Shanghai):

- ERS + DRS off: ≈ 3.6 s (TUM 94.89 → 91.24 s)
- AV-21 tyre transients: up to ≈ 8 s in closed loop. `sf70h_f1` fixes part of it.
- Track check uses the full footprint. TUM's raceline keeps the car on track for only 70 % of the lap.

Training curves (clean deterministic laps only)



Both runs plateau. Barcelona: best 83.97 s, then 84.0–84.7 s with only 26/142 clean deterministic evaluations. Shanghai: 101.3–102 s from 4M to 10.5M steps.

Open points

Summary and next steps

Summary

- The time-optimal reward (no jitter term, linear $|e|$ shaping as PBRS) lets the driver beat the QSS lap on its own reference line.
- The AV-21 tyre slip behaviour explains much of the gap between our car and an F1 car. `sf70h_f1` replaces it with F1-sourced values.
- Our OCP reproduces Limebeer's Barcelona lap within 0.3 %. RL is within 1.7 % of it (with a different car).

Decisions needed

1. Heave damping for `sf70h_f1`, or softer heave springs (porpoising).
2. I_z: keep 1159.6 (recommended) or use 560 kg m^2.
3. RL on `sf70h_f1`: scale env steering to 0.30 rad and re-tune demo traction-control targets.
4. Then: ERS on, a racing line sized for the car's footprint, and seeds 1–2.

Sources in the repository

- Reward: `docs/specs/R1_time_optimal.md`, `docs/research/reward_redesign.md`, `rlsetup/driving/config.py`, DECISIONS #13, #19
- Runs: `docs/results/{D2_full1, R1_warm, sf70h_mac_r1, bcn}`, `comparison/training_stop_20261029_094402Z/p1_TUM`
- AV-21 diagnosis: `comparison/tum_ocd_debug_20261031_144747Z/FINDINGS.md`, `docs/research/f1_car_parameters.md`
- New car: `docs/reports/SF70H_F1_UPGRADE.md`; ocd_first_lap branch `sf70h-f1` (`docs/SF70H_TUM_CAR.md`)
- OCP: `docs/results/limebeer_ocp/REPORT.md`, `docs/research/limebeer_lap_times.md`
- Real laps: `docs/research/shanghai_lap_records.md`, `shanghai_sim_and_car_reference.md`

Most of this material is on branches `comparison-tum-vs-rl` / `sf70h-f1`, not yet on `main`.